

Assignment - 6.3

2303a51144

Batch -17

Katkar Sanskar

Task1 : TO print student details includes roll no, name ,branch

Prompt :# Develop a simple student management module in Python using Object-Oriented Programming.

#The class should include attributes such as name, roll number, and branch.

#Add a method display_details() to print student information.

#Create sample student objects and display their details on the console.

#Ensure the code is clean, structured, and easy to understand.

#CODE

```
class Student:

    def __init__(self, name, roll_number, branch):

        self.name = name

        self.roll_number = roll_number

        self.branch = branch

    def display_details(self):

        print(f"Student Name: {self.name}")

        print(f"Roll Number: {self.roll_number}")

        print(f"Branch: {self.branch}")

# Sample object creation and output

student1 = Student("Alice", "101", "Computer Science")

student1.display_details()

student2 = Student("Bob", "102", "Mechanical Engineering")

student2.display_details()
```

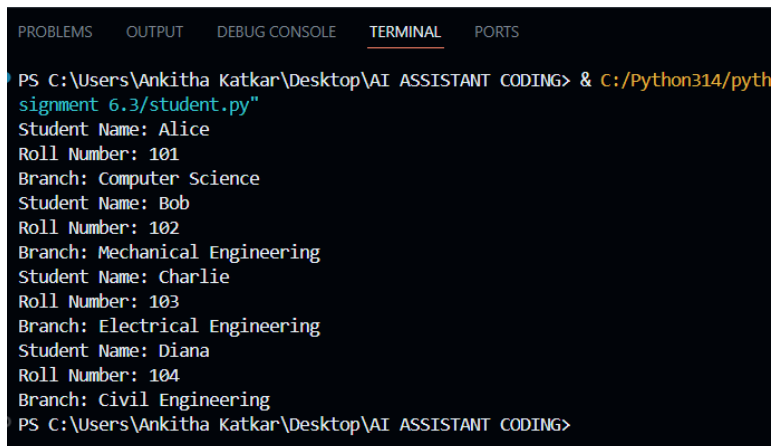
```
student3 = Student("Charlie", "103", "Electrical Engineering")

student3.display_details()

student4 = Student("Diana", "104", "Civil Engineering")

student4.display_details()
```

output :



```
PS C:\Users\Ankitha Katkar\Desktop\AI ASSISTANT CODING> & C:/Python314/python signment 6.3/student.py
Student Name: Alice
Roll Number: 101
Branch: Computer Science
Student Name: Bob
Roll Number: 102
Branch: Mechanical Engineering
Student Name: Charlie
Roll Number: 103
Branch: Electrical Engineering
Student Name: Diana
Roll Number: 104
Branch: Civil Engineering
PS C:\Users\Ankitha Katkar\Desktop\AI ASSISTANT CODING>
```

TASK – 2 To print first ten multiples of user input number

Prompt: #Write a Python function to print the first 10 multiples of a given number using a for loop.The program should be simple and easy to understand.give clean and structured code

CODE :

```
def print_multiples(number):

    print(f"The first 10 multiples of {number} are:")

    for i in range(1, 11):

        multiple = number * i

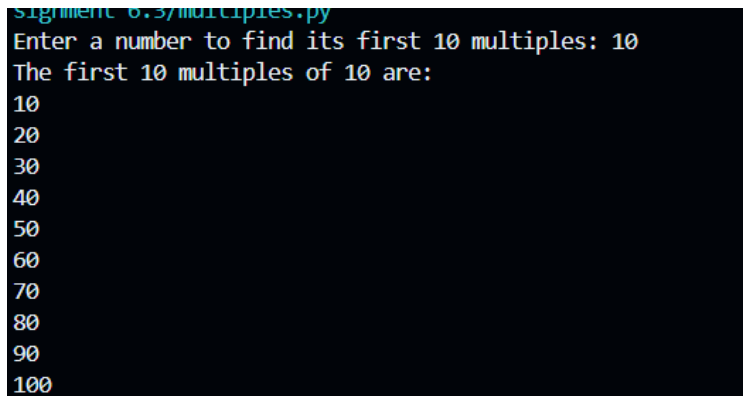
        print(multiple)

# using user input

num = int(input("Enter a number to find its first 10 multiples: "))
```

```
print_multiples(num)
```

OUTPUT :



```
Assignment 0.3/multiples.py
Enter a number to find its first 10 multiples: 10
The first 10 multiples of 10 are:
10
20
30
40
50
60
70
80
90
100
```

TASK – 3 : print that classifies age groups using if-elif-else conditional statements to classify age groups as child, teenager, adult, senior

PROMPT : #Write a Python function that classifies a person based on age using nested if-elif-else statements. The program should classify age into child, teenager, adult, and senior. Use simple conditions and clear logic. Give clean and structured code

CODE

```
def classify_age(age):
    if age < 0:
        return "Invalid age"
    elif age <= 12:
        return "Child"
    elif age <= 19:
        return "Teenager"
    elif age <= 59:
```

```
        return "Adult"

    else:

        return "Senior"

# using user input

age_input = int(input("Enter the age of the person: "))

classification = classify_age(age_input)

print(f"The person is classified as: {classification}")
```

OUTPUT :

```
segment 6.3/ageclassification.py"
Enter the age of the person: 25
The person is classified as: Adult
PS C:\Users\Ankitha Katkar\Desktop\AI ASSISTANT CODING> █
```

TASK- 4 :: You need to calculate the sum of the first n natural numbers.
Using for loop

PROMPT #Write a Python function named `sum_to_n()` that calculates the sum of the first n natural numbers using a for loop. The code should be simple and easy to understand. give clean and structured code

CODE :

```
def sum_to_n(n):

    total_sum = 0

    for i in range(1, n + 1):

        total_sum += i
```

```
    return total_sum

# using user input

num = int(input("Enter a natural number n to calculate the sum of the first n
natural numbers: "))

result = sum_to_n(num)

print(f"The sum of the first {num} natural numbers is: {result}")
```

OUTPUT :

```
Enter a natural number n to calculate the sum of the first n natural numbers: 10
The sum of the first 10 natural numbers is: 55
PS C:\Users\Ankitha Katkar\Desktop\AI ASSISTANT CODING> 
```

TASK-5 : You are designing a basic banking application

PROMPT : #Generate a # Write a Python program to create a BankAccount class.

The class should include methods deposit(), withdraw(), and check_balance().

Use simple logic and beginner-friendly code. Give clean and structured code.

CODE :

```
class BankAccount:

    def __init__(self, account_holder, initial_balance=0):

        self.account_holder = account_holder

        self.balance = initial_balance

    def deposit(self, amount):

        if amount > 0:
```

```
        self.balance += amount

        print(f"Deposited: ${amount:.2f}")
    else:
        print("Deposit amount must be positive.")
```

```
def withdraw(self, amount):
    if amount > 0:
        if amount <= self.balance:
            self.balance -= amount
            print(f"Withdraw: ${amount:.2f}")
        else:
            print("Insufficient balance.")
    else:
        print("Withdrawal amount must be positive.")
```

```
def check_balance(self):
    print(f"Current balance: ${self.balance:.2f}")
```

```
ram=BankAccount("John Doe", 1000)
```

```
ram.check_balance()
```

```
ram.deposit(500)
```

```
ram.check_balance()
```

```
ram.withdraw(200)
```

OUTPUT ;

```
Current balance: $1000.00
Deposited: $500.00
Current balance: $1500.00
Withdraw: $200.00
```